Course Name: Introduction to Python and Data Structures
Course Code: CS-121
Course Type: Institute Core

Contact Hours/Week: 2L/2P **Course Credits:**03

Course Objectives

- It aims to provide students with an understanding of the role computation can play in solving problems.
- To help students, regardless of their major, feel justifiably confident in their ability to write small programs that allow them to accomplish useful goals.

Unit Number	Course Content	Contact Hours
UNIT-01	Context of Software Development - Software - Learning Programming with Python, Values and Variables - Integer and String Values - Identifiers - User Input - String Formatting, Expressions and Arithmetic - Expressions - Arithmetic Examples.	08L
UNIT-02	Conditional Statements - Boolean expressions - If/Else statement - Other Conditional Expressions, Iteration – Loops, Using Functions - Introduction to Using Functions - Functions and Modules, Exceptions -try, except, else, pass, raise.	06L
UNIT-03	ng Functions -1 - Function Basics - Parameter Passing - Custom Functions vs Standard Functions – Refactoring, Writing Functions -2 - Global Variables - Making Functions Reusable - Functions as Data, Objects - Using Objects - String, File Objects.	06L
UNIT-04	- Using Lists - Building Lists - List Traversal, Tuples, Dictionaries, and Sets - Storing Aggregate Data - Enumerating the Elements of a Data Structure,	04L
UNIT-05	ılar Expressions, Class Design - Composition and Inheritance, classes, objects/instances, methods, inheritance, multiple inheritances, properties, decorators.	04L
UNIT-06	ductions to Data Structures, algorithm complexity, arrays, Linked List, Stack, Queues, Trees and Graphs (Memory Representation and a few basic operations)	07L

Course Outcomes

CO1: Understand basic principles of computers and understand basics of binary computation.

CO2: Understand the programming basics (operations, control structures, data types, etc.).

CO3: Apply various data types and control structure and understand class inheritance and polymorphism.

CO4: Understand the object-oriented program design and development and understand and begin to implement code.

Text Books

1. Guttag, John. Introduction to Computation and Programming Using Python: With Application to Understanding Data Second Edition. MIT Press, 2016.
2. Data Structures and Algorithms in Python by Roberto Tamassia.
3. Python Crash Course: A Hands-On, Project-Based Introduction to Programming by Eric Matthes, No Starch Press.

Reference Books

1. Automate the Boring Stuff With Python: Practical Programming for Total Beginners by Al Sweigart, No starch Press
2. Fluent Python: Clear, Concise, and Effective Programming by Luciano Ramalho, O'Reilly Media.
3. Python Cookbook by David Beazley, Brian K. Jones, O'Reilly Media, Inc.